EEN100 - Statistics and Machine Learning in High Dimensions

Project 2

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Sparse regression modeling

- In class, you have talked about sparse linear models and signal recovery
- For this project, we will focus on the regression setting and conditions under which we can achieve "sparsitency"
- The project is based on a highly cited paper by Zhao and Yu in Journal och Machine Learning Research 2008, though similar ideas were developed by other groups in parallel.

OLS regression

Let's start with the basic setup:

$$y = X\beta + \epsilon,$$

where y is $n \times 1$, X (the design matrix) is $n \times p$ and ϵ is a $n \times 1$ vector random variables constituting the additive errors.
We commonly used ordinary least squares to estimate the model parameters from data:

$$\arg\min_{\beta} ||y - X\beta||_2^2 \rightarrow -2X'(y - X\beta) = 0 \rightarrow X'X\beta = X'y$$

The last expression is called the *normal equations*.
IF $(X'X)$ is non-singular and can be inverted we can solve for β explicitly and obtain

$$\hat{\beta} = (X'X)^{-1}X'y$$

OLS regression ctd

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$E(\hat{\beta}) = \beta$$

$$V(\hat{\beta}) = \sigma^2(X'X)^{-1}$$

The key here is $X'X$!
If $p > n$, $X'X$ is singular and cannot be inverted. This is also true if there are perfectly linearly correlated columns in X.
If $p \sim n$, $X'X$ may be invertible but near singular with some very small eigenvalues.

$$V(\hat{\beta}) = \sigma^2(X'X)^{-1} = \sigma^2(VDU'UDV')^{-1} = \sigma^2 VD^{-2}V'$$

So if some eigenvalues of $X'X$ are small, the estimation variance explodes (unstable estimates).

Regularized regression

If $p > n$ or the inverse of $X'X$ is numerically unstable, it is common to use regularization techniques: e.g. $(X'X + \lambda I)$, essentially adding something to the diagonal of the matrix to stabilize the inverse operation.
We can arrive at the same proposal using penalized regression, constraining the parameters:

$$||y - X\beta||_2^2 + \lambda||\beta||_2^2,$$

called *ridge regression*.
This is no longer unbiased but can be asymptotically so if we allow for $\lambda_n \to 0$

Sparse regression

What if we want to stabilize the estimation process but also assume that the signal is, in fact, sparse, i.e. only a subset of βs are non-zero?

$$||y - X\beta||_2^2 + \lambda||\beta||_1,$$

is called the *lasso* and is an L1-regularized estimation procedure. This not only stabilizes the estimation but also provides sparse solutions.
Why is the solution sparse? The penalty term has non-continuous derivatives at 0 (jumps from -1 to $+1$), but subgradients $[-1, 1]$ - "gradient matching" results in sparse solutions.

Lasso - asymptotics

So does lasso always work? Can we retrieve the sparse solution?
Classical asymptotic results for OLS $\hat{\beta} \rightarrow_p \beta$, and for ridge if the ridge penalty goes to 0 as n goes to infinity.
Several papers studied lasso properties asymptotically by considering

$$y_n = X_n \beta^n + \epsilon_n,$$

where X_n now also grows in dimensionality (more features) as n grows, and the true model itself also depends on n (i.e. the model sizes may grow as the sample size grows. Still, β^n is assumed to be sparse.
Lasso solution: $\hat{\beta}^n = \arg\min_\beta ||y_n - X_n\beta||_2^2 + \lambda||\beta||_1$

Lasso asymptotics ctd

In the simple case of fixed model size and fixed dimensionality, Knight and Fu (2000) showed that you get consistent parameter estimates using lasso provided that

- $\lambda_n = o(n)$
- and $C_n = \frac{1}{n}\sum_i x_i x_i^T \rightarrow C$
- $\frac{1}{n}\max_i x_i x_i^T \rightarrow 0$

Lots of papers were published on lasso asymptotics and finite properties. It was noted that the covariance structure of the features was key to model selection using lasso being model consistent.

Irresprentability condition

Zhao and Yu investigate the more complex setting with dimensionality and true model size depending on n.
The main results in the paper connect model consistency of lasso to the so-called irrepresentability condition.

- What if there are correlations between true predictors? Well, up to a certain level this can be handled with the so-called *elastic net* - using a combination of the ridge and lasso penalties. Then, correlated features tend to be selected together in the model.
- What if there are correlations between true and false predictors? Well, clearly now we may have a challenge selecting the true model. Elastic net is a stabilizing approach but will lead to the inclusion of many false features in your model.

Sparsistency

Consistent estimators: $\hat{\beta}^n \rightarrow_p \beta^n$
Sparsistency: $P(\{i : \hat{\beta}_i^n \neq 0\} = \{\text{i: } \beta_i^n \neq 0\}) \rightarrow 1$ as $n \rightarrow \infty$
Ideally, we want both.
In the paper Zhao and Yu utilize an even stronger consistency: sign consistency.

$$sign(\hat{\beta}^n) = sign(\beta^n)$$

with the sign operator taking on values -1, +1 and 0.
If we select the correct zero-valued βs as well as the correct sign for the non-zero βs we denote this as

$$\hat{\beta}^n =_S \beta^n$$

Strongly consistent

Lasso is *strongly consistent* if

$$lim_{n\to\infty} P(\hat{\beta}^n(\lambda_n) =_S \beta^n) = 1$$

for some $\lambda_n = f(n)$ that does not depend on y_n, X_n.
That is, there is a way to pick a λ a priori such that lasso works.

The weaker result, general consistency, states that there is a λ for which lasso will select the correct model for each realization of data - meaning you have to tune this hyperparameter, as you would do in practical settings.

Irrespresentability, ctd

All centers of the feature covariance matrix $C^n = \frac{1}{n}(X^n)'X^n$.
Let us assume that the first q features in X^n correspond to the non-zero βs and the p-q remaining features are not part of the true model.
Let us denote the partition of $X^n = [X^n(1)X^n(2)]$ to separate the relevant from the irrelevant features.
We can now write the feature covariance matrix as

$$C^n = \begin{pmatrix} C^n_{11} & C^n_{12} \\ C^n_{21} & C^n_{22} \end{pmatrix}$$

Irrespresentability, ctd

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- Strong irrepresentability condition:
  $\exists \eta \ s.t. \ |C^n_{21}(C^n_{11})^{-1} sign(\beta^n_{(1)})| \leq 1 - \eta$
- Weak irrepresentability condition:
  $|C^n_{21}(C^n_{11})^{-1} sign(\beta^n_{(1)})| < 1$

Irrespresentability, ctd

- Strong irrepresentability condition:
  $\exists \eta \ s.t. \ |C_{21}^n (C_{11}^n)^{-1} sign(\beta_{(1)}^n)| \leq 1 - \eta$

What does this mean?
Let's consider regression of the irrelevant features $X^n(2)$ on the relevant ones, $X^n(1)$ with OLS:

$$\hat{\beta}_{2on1} = (X^n(1)'X^n(1))^{-1} X^n(1)'X^n(2) = (C_{11}^n)^{-1} C_{12}^n = C_{21}^n (C_{11}^n)^{-1}$$

If this should be less than $1 - \eta$, we need to have a constraint on the regression coefficients in model $1 \rightarrow 2$, i.e. limited correlation between the two feature sets, or that the irrelevant features can't be "represented" by the true features.

Irrespresentability, ctd

- Strong irrepresentability condition:
  $\exists \eta \ s.t. \ |C_{21}^n (C_{11}^n)^{-1} sign(\beta_{(1)}^n)| \leq 1 - \eta$

The sign also matters! Let's say that all features are positively correlated but the β_1, β_2 have opposite sign. This means that the "signal" in the features related to y has to live in the uncorrelated part of x_1, x_2. Lasso can therefore "tell" that x_3 is not in the model.

Irrespresentability, ctd

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The explanation above was oversimplified. However, consider the case where x_1, x_2 are correlated ρ_1 and the correlated between x_3 and x_1, x_2 is ρ_2.
$C_{21}^n (C_{11}^n)^{-1} = (\frac{\rho_2}{1+\rho_1}, \frac{\rho_2}{1+\rho_1})$ which is additive if the $sign(\beta)$ same signs and can lead to a violation of the condition but *cancels out!!!* if the signs are opposite.

Irrespresentability, ctd

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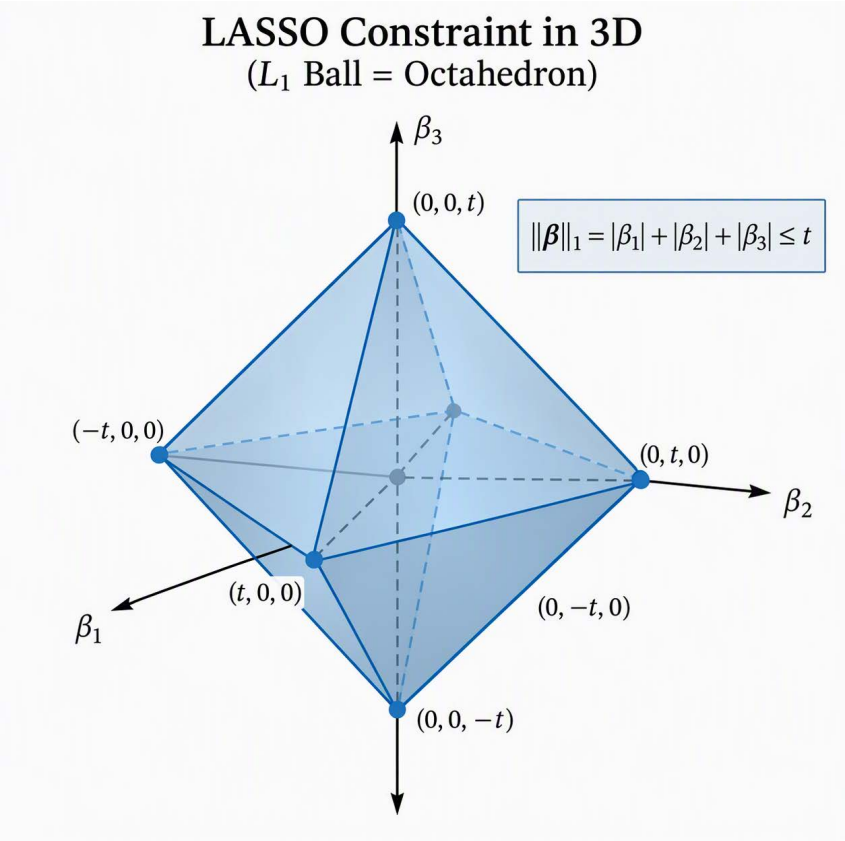
Even more simple - consider the geometry of the problem. If the features are all positively correlated, the least squares solution is essentially $\beta_1 + \beta_2 + \beta_3 = constant$ (illdetermined) which looks like an elongated cylinder, almost a sheet, in 3D. In the β_1, β_2 plane we have an elongated ellips tilted -45 degrees, and this is true for every level plane $\beta_3 = c$ as well. The cylinder tilt toward the β_3 axis as we move up the planes, parallell to the double-pyramid (3D lasso constraint) side in the positive quadrant.

Irrespresentability, ctd

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  $\exists \eta \ s.t. \ |C_{21}^n (C_{11}^n)^{-1} sign(\beta_{(1)}^n)| \leq 1 - \eta$

The cylinder "tilt" toward the β_3 axis as we move up the planes, parallell to the octahedron (3D lasso constraint) side in the positive quadrant.
If the true β_1, β_2 have the same sign, this is the worst possible case - the cylinder tilts toward β_3 as we move up the plane and it is likely β_3 will be selected.
If the βs have opposite signs, the cylinder "tilts" away from the β_3 axes and it is unlikely the noise feature will be selected.

Irrespresentability, ctd



Proofs in the paper

The proofs for sign consistency are based on the probability of two set events being shown to converge to 1.
One event primarily deals with the estimation consistency for the true non-zero coefficients and the other with limiting the inclusion of irrelevant predictors.
Theorem: If strong irrepresentability condition holds with $\eta > 0$

$$P(\hat{\beta}^n(\lambda_n) =_S \beta^n) \geq P(A_n \cap B_n)$$

- $A_n : |(C_{11}^n)^{-1} W^n(1)| < (\sqrt{n}|\beta_{(1)}^n| - \frac{\lambda_n}{2n}|(C_{11}^n)^{-1} sign(\beta_{(1)}^n)|)$
- $B_n : |C_{21}^n (C_{11}^n)^{-1} W^n(1) - W^n(2)| \leq \frac{\lambda_n}{2\sqrt{n}} \eta$

where $W^n(j) = X^n(j)^T \epsilon_n / \sqrt{n}$

Lasso works when?

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where $W^n(j) = X^n(j)^T \epsilon_n / \sqrt{n}$
When λ_n decreases (weak constraint), then odds of event A_n (finding the true coefficients with correct sign) increases but B_n decreases (risk of false inclusions) - so trade off between bias (sign consistency of true features) and selection accuracy,
Larger η leads to larger B_n, protection against false positives, but does not affect A_n

Lasso works when?

So, the first event is about consistency, and the other about false inclusions - which is where correlation structure comes into play as a key constraint.
What about the details?
The projections onto the residuals $(X^n)^T \epsilon_n$, with proper scaling, opens up for standard convergence proofs to be utilized (sums of standardized rv),
All together, it boils down to the correlation structure between the relevant features and the relevant-irrelevant features.

Lasso works when?

You don't have to read the proof in full - the key part of the project is to reproduce the results of the theorem, not derive the proof of it.
A_n related to the signal strength in the model - given the λ_n (how much we constrain the fit), is the signal strong enough that the active feature coefficient estimates don't cross 0 and change sign of the true coefficients?
B_n essentially checks that the irrelevant features aren't too correlated with the residuals so that they remain inactive (given λ_n - constraint region).

Main results

In the low-dimensional case, there are already results from previous papers,
However, in this paper, the authors utilize the setup to state that

$$P(\hat{\beta}^n(\lambda_n) =_S \beta^n) = 1 - o(e^{-n^c}), 0 \leq c < 1$$

That is, sparsistency has exponential decay in the low-dim paradigm.

Main results, ctd

What about the high-dimensional case? (NOTE FOR PROJECT - high-dimensional means $p = p_n$, the number of features grow with n, not just that p is large!!!)
Here, we need a slightly more stringent assumption on the errors. Let us allow for dimensionality and model complexity to grow with n. To make the previous results carry over we need to control the size of the smallest relevant coefficient and the correlation between the relevant features (sounds a bit familiar from the previous project?).
Without repeating explicitly from the paper; what do we need?

- Well-behaved features (moments)
- Eigenvalues bounded from below of C_{11}^n: i.e. if you knew the true model - is the problem doable?
- $q_n = O(n^{c_1})$ - sparsity assumption
- $n^{\frac{1-c_2}{2}} \min_i |\beta_i^n|| \geq 0$ - the smallest relevant coefficient can't be too small

where $0 \leq c_1 \leq c_2 \leq 1$

Main results, ctd

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The penultimate and ultimate conditions together - has to do with signal strength vs noise level impact on estimation variability $(\sqrt{n})^{-1}$

Main results, ctd

For the large q, large p, under the conditions from above and for well behaved errors (finite k moments), we have the following result:
For $\frac{\lambda_n}{\sqrt{n}} = o(n^{\frac{c_2-c_1}{2}})$ and $\frac{1}{p_n}(\frac{\lambda_n}{\sqrt{n}})^{2k} \to \infty$

$$P(\hat{\beta}^n(\lambda_n) =_S \beta^n) \geq 1 - O(\frac{p_n n^k}{\lambda_n^{2k}}) \to_{n\to\infty} 1$$

A polynomial convergence in the high-dimensional case (vs exponential in the fixed, low-dimensional case). With gaussian noise, we also get exponential convergence. So, we can have p grow faster than n and still obtain model selection consistency.

Simulations and practical results

Finally, in the paper they also discuss if the irrepresentability conditions required to hold for model selection consistency are trivial.
In the high-dimensional case it is not. Spurious correlations between features can create problems. Zhao and Yu present some practical results linking irrespresentability to bounds on pairwise correlations between features and study some commonly used simulation setups.

The Project

- Read the paper - focusing on the irrepresentability condition and the theorems 1 and 3 (the main convergence results)
- Simulate your own version of the low-dimensional setting, showing when lasso works and fails due to the irrepresentability condition not holding
- Create a high-dimensional setting and simulate in order to study optimal λ as a function of n for at least 3 different settings of p, q and correlation structure. Check what η is in your setting and make sure you explore "easy" and "difficult" cases. [1]
- Present your main results on ~ 5 slides.

[1] Caution: in the paper the lasso problem is defined as $||Y_n - X_n\beta_2^2 + \lambda||\beta||_1$ whereas many implementations of lasso use $\frac{1}{n}||Y_n - X_n\beta_2^2 + \tilde{\lambda}||\beta||_1$, meaning $\lambda = n\tilde{\lambda}$